



Instruction for Use TA-Implant-System

1. Intended Use

This instruction manual applies to all components of the TA Implant System, including TA-Switch / Plus implants and corresponding components.

Each component is identified by the article number (REF) and description on the packaging label.

2. General Instructions

- Only qualified and experienced professionals should perform implant procedures.
- Clinicians must have proficiency in surgical techniques and the ability to evaluate normal and pathological treatment outcomes.
- Preoperative prosthetic planning is mandatory and should be completed before implantation. If surgical conditions require adjustments, modifications can be made accordingly.
- This instruction manual alone is not sufficient for the immediate application of the TA Implant System. Training and experience in implantology are essential.

3. Product Description

The TA Implant System is designed for endosteal dental implant procedures, used in oral surgery to restore function and aesthetics in partially or fully edentulous patients.

- The system consists of:
- Titanium alloy implants with a modified surface for optimal osseointegration.
 - Surgical instruments required for precise implant placement.
 - Prosthetic components designed for final restoration.

The system is suitable for both single-stage and two-stage implant procedures.

4. Material Composition

Implants: Manufactured from Grade 4 pure titanium.

Abutments and screws: Made from Grade 5 titanium alloy (Ti-6Al-4V).

Surgical drills: Manufactured from high-quality surgical steel.

Plastic components: Marking guides are made from polyethylene.

5. Indications

- The TA Implant System is indicated for:
- Fixed or removable prosthetic restorations in fully or partially edentulous jaws.
 - Single-tooth replacement with implant-supported crowns.
 - Anchorage for orthodontic appliances in cases requiring skeletal support.

Sufficient bone volume, density, and height are necessary for successful implantation.

6. Contraindications

Absolute Contraindications

- TA implants should not be used in patients with:
- Uncontrolled systemic diseases (e.g., severe diabetes, osteoporosis).
 - Recent or ongoing radiation therapy affecting the implant site.
 - Psychological disorders or chronic pain syndromes.
 - Poor oral hygiene that may compromise healing.
 - Allergy to titanium or implant materials.

Relative Contraindications

Special precautions should be taken in patients with:

- Controlled diabetes.
- Bruxism (teeth grinding).
- Smoking habits (which may impair osseointegration).
- Alcohol or drug abuse.
- Reduced bone volume: If anatomical structures (e.g., nerves, sinuses) are too close, alternative prosthetic options should be considered.

7. Preoperative Assessment

A comprehensive assessment should be performed, including:

- Medical history review to identify risk factors.
- Clinical examination of soft tissues and occlusion.
- Radiographic imaging (panoramic X-ray, CBCT if necessary) to evaluate bone structure.
- Discussion of treatment alternatives and patient education about post-treatment care.

The patient must provide informed consent, acknowledging the risks and benefits of implant treatment.

8. Possible Side Effects

Temporary Side Effects

- Pain, swelling, mild bleeding.
- Temporary difficulty in speech or chewing.

Potential Complications

- Bone resorption (horizontal and vertical bone loss) may occur over time.
- Nerve damage if implants are placed too close to critical structures.
- Infection and peri-implantitis if oral hygiene is inadequate.
- Failure of osseointegration, leading to implant loss.
- Chronic pain, although rare, has been reported in some cases.

To minimize risks, antibiotic prophylaxis is recommended before surgery.

9. Surgical Protocol

The following surgical guidelines must be observed:

- Use rotary and displacement instruments under local anesthesia.
- Drilling sequence should follow TA Implant System guidelines for optimal bone preparation.

Recommended Drilling Sequence (TA Switch / Plus Implants)

- TA-3.3: DR 2.0 → 2.8 → CS 3.3 → T 3.3
- TA-3.8: DR 2.0 → 2.8 → 3.2 → CS 3.8 → T 3.8
- TA-4.1: DR 2.0 → 2.8 → 3.2 → 3.5 → CS 4.1 → T 4.1
- TA-4.8: DR 2.0 → 2.8 → 3.2 → 3.5 → 4.2 → CS 4.8 → T 4.8
- TA-5.5: DR 2.0 → 2.8 → 3.2 → 3.5 → 4.2 → 4.9 → CS 5.5 → T 5.5

(DR = Drill; CS = Countersink; T = Tap)

The specified lengths of the implants refer only to the length of the rough surface. With the addition of the polished surface at the neck of the implant, the total length is actually longer than the specified length. The polished surface is approximately 0.3 mm in height for 3.3 Passive Implants, whereas for all other implant types and diameters, it is approximately 0.45 mm in height.

The length markers on the drills indicate a length that is longer than the implant length, measured from the tip of the drill. The length markers on the drills are located at 6.5, 8.5, 10.5, 12.5, and 14.5 mm from the tip.

An appropriate safety margin must be maintained to protect adjacent anatomical structures.

During bone preparation, the implant site must be continuously cooled using pre-cooled physiological saline solution.

Drilling speed must not exceed 500-800 RPM, and only sharp drills should be used, with a maximum of 10 uses per drill.

The intermittent milling and drilling technique is recommended to prevent excessive heat generation.

The implant should be removed from the sterile packaging immediately before placement and inserted into the prepared implant site using the pre-installed insertion aid.

Once positioned, the implant should be seated into the bone at the desired depth using the implant key.

In cases of high bone resistance, a thread cutter may be used to create the necessary space.

After placement, the implant should be sealed either with a cover cap (two-stage procedure) or a healing screw of appropriate length (single-stage procedure).

Adequate wound management and, if necessary, a radiographic evaluation should be performed to ensure proper healing.

During the healing period, which typically lasts 3-6 months depending on bone quality, the surgical site should be monitored at regular intervals.

If incomplete soft tissue healing or premature exposure of the implant occurs, replacing the cover cap with a healing screw is recommended.

If the mechanical stability of the implant and the clinical situation at the time of placement allow, immediate loading of the implant may be considered. This decision is at the discretion of the treating dentist.

In cases of uncertainty, a healing period of 3-5 months without loading is recommended, depending on the bone quality.

At the end of the healing process, radiographic and clinical evaluations must be conducted to confirm osseointegration.

Prosthetic components of the TA Implant System are available for final restoration after successful healing.

10. More Clues:

Implants must only be used within the expiration date stated on the package.

Implants with damaged sterile packaging must not be used and should be returned to the manufacturer.

After implantation, the implant type and manufacturing batch number (LOT-number on the label) must be recorded on the patient's medical file.

Implants must be stored in a dry, sealed environment. The packaging should only be opened immediately before insertion.

Any contact with foreign substances prior to insertion must be strictly avoided.

The endosteal part of the implant must not be touched to maintain sterility.

If an implant, abutment, or any other material is accidentally swallowed, the object must be located using radiographic imaging (X-ray, CT, etc.), and appropriate medical intervention must be carried out immediately.

The healing progress of implants must be regularly monitored (e.g., via radiographic examination).

To prevent bone loss, any non-osseointegrated or infected implants must be removed in a timely manner.

The decision on the removal timeframe is at the discretion of the treating dentist.

11. Permanency and Instructions on Sterilization:

If the validity of specific components in the TA Implant System is limited, the expiration date will be clearly indicated on the label (hourglass symbol).

The TA Implant System components, including TA-Switch and TA-Plus implants, are supplied as sterile single-use products and sterilized using gamma radiation.

Cover caps, healing screws, and surgical instruments are delivered non-sterile in unautoclaved packaging and must be sterilized prior to use using standard sterilization procedures in dental practice.

We recommend sterilization at 120°C for 15 minutes for cover caps and healing screws.

For surgical instruments, we recommend 134°C for 8 minutes, followed by 5 minutes of cooling time.

Further cleaning and sterilization guidelines can be found on our official website.

12. Legend of the label:

	Sterilised using irradiation
	Batch Code
	Date of manufacture
	Manufacturer
	Do not re-use
	Use before expiry date
	Article number
	Serial number
	Consult instructions for use
	Unsterile

13. Form of Delivery

TA-Switch / Plus Passive Screwing Implants:

- Internal Hex:
 - SW 2.1 (for all 3.3 mm, 3.8 mm and 4.1 mm implants, as well as 4.8 x 6 mm and 5.5 x 6 mm implants)
 - SW 2.5 (for 4.8 mm and 5.5 mm implants)
- Diameters and Lengths:
 - Ø 3.3 mm: 8, 10, 12, 14 mm
 - Ø 3.8 mm: 6, 8, 10, 12, 14 mm
 - Ø 4.1 mm: 6, 8, 10, 12, 14 mm
 - Ø 4.8 mm: 6, 8, 10, 12, 14 mm
 - Ø 5.5 mm: 6, 8, 10, 12, 14 mm

TA-Switch / Plus Aggressive Screwing Implants:

- Internal Hex:
 - SW 2.1 (for 3.3 mm, 3.8 mm and 4.1 mm implants)
 - SW 2.5 (for 4.8 mm and 5.5 mm implants)
- Diameters and Lengths:
 - Ø 3.3 mm: 8, 10, 12, 14 mm
 - Ø 3.8 mm: 8, 10, 12, 14 mm
 - Ø 4.1 mm: 8, 10, 12, 14 mm
 - Ø 4.8 mm: 8, 10, 12, 14 mm
 - Ø 5.5 mm: 8, 10, 12, 14 mm

Implants are delivered sterile. The lengths refer to the entire lengths of the implants. Deliverable products of the TA-implant-system may be found in the current product catalogue or rather the current price list or on the website of the TA Zahnimplantate GmbH

TA Zahnimplantate GmbH reserves its right to change the design of the products and components and their package, to adjust application notes and to remake prices and delivery conditions. The accountability is restricted to the replacement of faulty products. Further claims of any kind are excluded.



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